holding money becomes more costly compared to earning potential returns from other investments.

When the demand for money rises, it leads to higher interest rates. Higher interest rates can dampen investment and consumer spending, reducing aggregate demand. This reduction in aggregate demand can slow down economic growth and potentially lead to a recession if not addressed promptly. We will explore the concept of demand for money in greater details in Unit 5: Demand for Money.

To counteract the negative impact on aggregate demand, the central bank can increase the money supply. Aggregate demand rises when money supply increases faster than money demand — with a concomitant rise in output or the price level.

By injecting more money into the economy, the central bank can stabilise interest rates and support aggregate demand. This can happen through mechanisms like open market operations, where the central bank buys government securities, thereby increasing the reserves of commercial banks and enabling them to lend more. We will discuss more about various instruments of monetary policy in Block 3.

An increase in the money supply typically lowers interest rates, making borrowing cheaper for businesses and consumers. Lower interest rates reduce the cost of financing investments and major purchases, leading to higher spending on capital goods and consumer products. As businesses expand and consumers increase their expenditure, aggregate demand rises, boosting economic output and employment. However, if the economy is already near full capacity, the increased demand may primarily lead to higher prices, causing inflation.

This intervention requires timely recognition of changes in money demand to avoid economic instability. If the monetary authority fails to recognize and respond to this shift promptly, the economy may experience slower growth or even a contraction.

The Transmission Mechanism of Monetary Policy: The Credit Channel

Ben Bernanke's seminal work "Credit in the Macroeconomy" explores the critical role of credit in influencing economic activity and macroeconomic stability. Bernanke's analysis highlights how fluctuations in credit availability can significantly impact economic performance, often through channels that traditional monetary theories might not fully account for.

Bernanke emphasizes the importance of the "credit channel" of monetary policy, which operates alongside the traditional interest rate channel. The credit channel affects the economy through the availability and terms of credit, impacting businesses' and consumers' ability to borrow and spend. This is especially relevant during periods of financial distress when credit markets tighten, leading to a reduction in investment and consumption, and consequently, a slowdown in economic activity.

Bernanke and Blinder (1988) highlighted how monetary policy impacts bank lending behaviour, establishing a foundational understanding of the bank lending channel. Bernanke and Gertler (1995) explored the balance sheet channel, showing how changes in firms' net worth influence their investment decisions and the broader economic impact. Recent studies have shown that during the 2008 financial crisis, the impaired credit channel significantly contributed to the depth and duration of the recession (Ivashina and Scharfstein, 2010).

The credit channel is particularly relevant in the presence of asymmetric information in financial markets. Banks and lenders have more information about the creditworthiness of borrowers than external investors, making bank lending a crucial source of finance for many firms and households. (Bernanke, 1993)

However, it is also important to note that the effectiveness of the credit channel can vary over the business cycle. During economic downturns, even with expansionary policy, banks may be hesitant to lend due to increased risk perceptions, limiting the channel's effectiveness. Conversely, during booms, easier credit can exacerbate economic expansions.

The credit channel can be broken down into two main components:

The Bank Lending Channel

The Balance Sheet Channel

Both components emphasize the role of financial intermediaries (banks) and the financial health of borrowers in the transmission of monetary policy.

1) **The Bank Lending Channel**

The bank lending channel operates through the impact of monetary policy on banks' ability to lend. We have already discussed it earlier how when the Central Bank implements expansionary monetary policy (e.g., lowering the interest rate or purchasing securities), it increases the reserves of commercial banks. Then, the banks have more funds available to lend. As the cost of funds decreases, banks can offer more loans at lower interest rates. With more accessible and cheaper credit, businesses and consumers are more likely to borrow for investment and consumption. This increased borrowing can lead to higher aggregate demand, stimulating economic growth. Conversely, contractionary monetary policy (e.g., raising interest rates) reduces bank reserves, leading to tighter lending conditions, higher borrowing costs, and lower aggregate demand.

2) **The Balance Sheet Channel**

The balance sheet channel focuses on how monetary policy affects the financial position of borrowers. Expansionary monetary policy often leads to higher asset prices (e.g., stocks, real estate). As asset values increase, the net worth of businesses and households improves. Higher net worth enhances the value of collateral that borrowers can pledge, improving their borrowing capacity. With better financial health, borrowers face lower borrowing costs and more favourable loan terms. Improved balance sheets make businesses more willing to invest in new projects and consumers more inclined to spend, further boosting economic activity. In a contractionary scenario, falling asset prices reduce net worth and collateral values, restricting borrowing capacity, and dampening investment and consumption.

Check Your Progress 3

1) Briefly describe the relationship between money, credit and macroeconomic policy.

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2) Explain the credit transmission channel of monetary policy.

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4.6 LET US SUM UP

The unit provided a discussion about the definition, nature and functions of what is commonly called 'money'. We saw that anything that is universally accepted as a medium of exchange is called money. The unit briefly discussed the disadvantages of a barter economy, and how the introduction of money facilitates exchange and reduces transaction costs. The unit also explained the meaning of the concept of 'liquidity'. We saw money is the most 'liquid' of instruments, and some other instruments, which perform some of the same functions that money does, are not as 'liquid' as money and hence are called 'near monies.

The unit discussed the functions of money. We saw that money serves as a medium of exchange, a unit of account, a store of value, and a standard of deferred payments. This last function which allows a transaction and its payment to be separated in time, is the source of credit and debt in the economy. The unit brought out the difference between money, income, wealth and assets. The system of payments in a modern economy was discussed and the means and methods of payments were listed. Elements of the macroeconomic relationship between money and credit and various macroeconomic variables were discussed.

4.7 KEY WORDS

Barter : A system where goods and services are exchanged for other goods and services.

Near Money : Highly liquid assets which are not cash but can easily be converted into cash.

Cryptocurrency : Digital or virtual currencies that use cryptography for security and operate on decentralized networks based on blockchain technology.

Monetary Policy : Monetary policy is the policy adopted by the monetary authority of a nation to affect monetary and other financial conditions to accomplish broader objectives like high employment and price stability.

Digital Currency : Digital currency is any currency, money, or money-like asset that is primarily managed, stored or exchanged on digital computer systems.

Unified Payments Interface (UPI) : Unified Payments Interface (UPI) is a real-time payment system in India that enables seamless money transfers from one bank account to another instantly and free of charge through a mobile device.

Real Time Gross Settlement (RTGS) : Real-time gross settlement systems are specialist funds transfer systems where the transfer of money or securities takes place from one bank to any other bank on a "real-time" and on a "gross" basis to avoid settlement risk.

National Electronic Funds Transfer (NEFT) : The NEFT or National Electronic Funds Transfer is an electronic payment system that allows users to initiate direct one-to-one payment anywhere across the country. One can send money to the beneficiary only if he or she has a bank account with any branch in the country.

4.8 SOME USEFUL REFERENCES

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4.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS

Check Your Progress 1

1) Refer to Section 4.2

2) Refer to Section 4.2

3) Refer to Section 4.2

Check Your Progress 2

1) Refer to Section 4.4

2) Refer to Section 4.4

3) Refer to Section 4.4

Check Your Progress 3

1) Refer to Section 4.5

2) Refer to Section 4.5

UNIT 5 DEMAND FOR MONEY

Structure

5.0 OBJECTIVES

The objectives of the units are as follows:

- Learning the meaning of demand for money;
- Understanding the main factors affecting the demand for money; and
- Understanding various economic theories of demand for money.

5.1 INTRODUCTION

Money demand plays a crucial role in understanding the broader functioning of economies, influencing monetary policy decisions, and shaping financial markets. We start our exploration with the meaning of money demand curve and the multifaceted factors that influence money demand, from interest rate to technological advancements. These factors lay the groundwork for our deep dive into various theories of money demand, each offering unique insights into this fundamental aspect of economic behaviour. First, we will explore the classical Quantity Theory of Money followed by the Cambridge version. Next, we will delve into Keynes' Liquidity Preference Theory, which emphasizes various motives of holding money and the role of interest rates in shaping individuals' decisions to hold money versus other assets. We will also examine the Baumol-Tobin theory for transaction demand, which focuses on the transaction motive for holding money balances and introduces the concept of opportunity cost in deciding between holding money and holding interest-bearing assets. Furthermore, we will explore Tobin's theory of speculative demand for money, which sheds light on how the portfolio of cash and bonds is optimized and its relationship with the speculative demand for money. Finally, the unit explores the Friedman's quantity theory of money. By the end of this unit, you will gain a comprehensive understanding the demand for money equipping you with valuable insights into the functioning of an economy.

5.2 MONEY DEMAND

Money demand means the quantity of money (or cash) which households and firms want to hold with themselves, but not in other forms like bonds. Keeping other factors constat, just as the demand for a good is negatively related with its price, similarly, the demand for money is negatively related with interest rate. Therefore, a money demand curve (showing the relationship between interest rate and the money demand) slops downward as shown in Figure 5.1. The simple reasoning behind this negative slope is that holding money at home does not earn anything and it loses the interest being offered by other assets like government bonds if that money is invested in the bonds. Thus, interest rate is the opportunity cost of holding or demanding money.

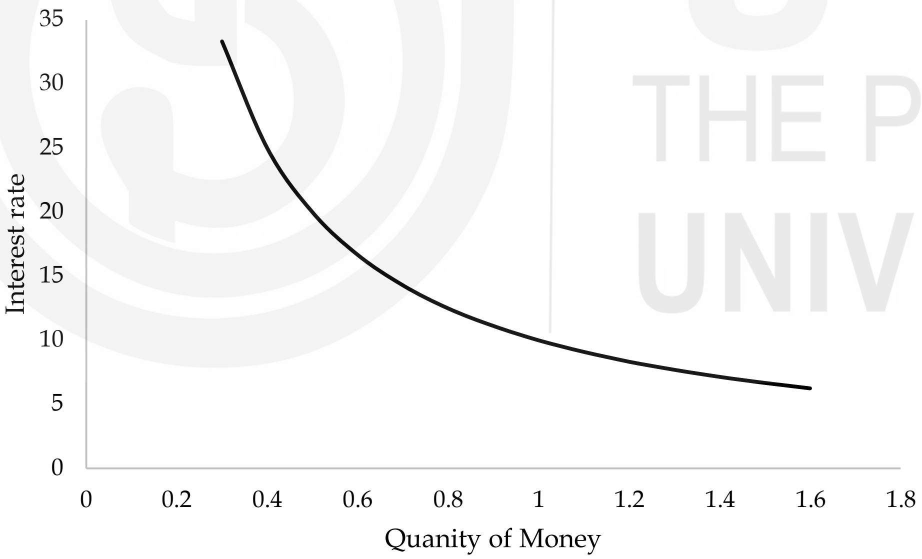

Fig. 5.1: Money Demand Curve

Check Your Progress 1

1) What is money demand?

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2) Explain the money demand curve.

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5.3 FACTORS AFFECTING THE DEMAND FOR MONEY

The main factors affecting the demand for money are as follows:

1) Interest rate

Keeping other factors constant, the demand for money is influenced by the anticipated returns of both money and alternative assets. When the expected return on money rises, the demand for money increases. Conversely, when the expected return on alternative assets increases, individuals with wealth tend to shift from holding money to investing in these higher-return alternatives, resulting in a decrease in the demand for money. For example, suppose that an investor who has a certain amount of wealth to allocate between holding money and investing in stocks. Suppose initially, the expected return on both holding money (such as keeping it in a savings account) and investing in stocks is 5%.

Now, let us suppose that there is a shift in the market, and the expected return on stocks increases to 8%. In this scenario, the investor may decide to reallocate some of their wealth from holding money to investing in stocks. This decision is driven by the desire to earn a higher return on their investment.

Conversely, if the expected return on holding money increases to 7%, while the expected return on stocks remains at 5%, the investor might decide to hold more money rather than investing in stocks. This is because the higher return on holding money makes it a more attractive option compared to investing in stocks.

In both cases, the investor's decision to allocate their wealth between holding money and investing in alternative assets is influenced by the expected returns of both options, illustrating the principles outlined in the theory of portfolio allocation.

2) Price level

When prices rise, people require more currency (e.g., rupee or dollar) to carry out transactions, leading to a greater desire to hold onto money. Let us consider an example: Imagine a small island economy where the price level for basic goods like fruits and vegetables is very low compared to a modern city. In this island economy, let's say a basket of fruits costs only a few island dollars. Now, fast forward a few decades, and due to economic development and inflation, the price level on the island has increased significantly. That same basket of fruits now costs several times more island dollars. As a result of this price increase, the islanders find themselves needing to hold more island dollars to conduct their daily transactions. Their nominal demand for money increases because they require a larger quantity of currency to purchase the same goods and services as before. This increase in the price level directly impacts the amount of money people wish to hold, highlighting the proportional relationship between the price level and the nominal demand for money.

3) Income (or GDP)

Households and businesses retain money to streamline their transactions for goods and services. As their purchasing activity expands, so does their desire to hold more money at any specific interest rate. Consequently, a rise in real GDP, reflecting the overall quantity of goods and services exchanged in the economy, changes the money demand.

4) Risk Associated with Alternative Assets

If the risk associated with alternative assets like stocks and real estate significantly rises, individuals may seek safer options, including holding onto money. Consequently, an increase in overall economic risk may lead to a higher demand for money. Suppose there is a period of economic instability where the stock market experiences a sharp decline, and there is uncertainty regarding the future performance of real estate investments. During such times, investors might perceive stocks and real estate as riskier assets due to the volatility and uncertainty in the market. In response to this increased risk in alternative assets, individuals may choose to shift their investment strategy towards safer options. They might decide to hold onto more cash or invest in low-risk assets like government bonds or savings accounts, which offer stability and security during uncertain times. This increased preference for safer assets, including holding onto money, illustrates how changes in the riskiness of alternative investments can influence the demand for money.

5) Liquidity of Alternative Assets

As alternative assets become more readily convertible into cash, the necessity of holding money diminishes. Similarly, if the liquidity of alternative assets is less, then the demand for money increases.

6) Payments and Receipts Technology

UPI (Unified Payments Interface) platforms such as PhonPe, Google Pay, Paytm have substantially reduced the demand for physical cash in India. Similarly, net banking system of payments and receipts have also reduced the need to carry money in hands. Offering instant, secure transactions directly from bank accounts via smartphones, they provide unparalleled convenience. Moreover, UPI's accessibility extends even to remote areas with limited banking infrastructure. Enhanced security features instill confidence in users, while incentives like cashbacks further incentivize digital transactions. Integration into various businesses and services ensures widespread acceptance, encouraging consumers to opt for digital payments over cash. Consequently, UPI has transformed India's payment landscape, making digital transactions the norm and significantly decreasing reliance on physical currency.

5.4 THEORIES OF MONEY DEMAND

Over a period of time various theories explaining money demand have been developed. These theories are explained here:

5.4.1 Quantity Theory of Money

Although, the quantity theory of money is basically about the money supply not the demand for money, but it can be used to find out the demand for money in equilibrium when the demand for money is equal to its supply. Therefore, we first understand the quantity theory and then will relate it to the demand side.

According to N.G Mankiw, David Hume is credited with the development of the quantity theory of money. However, American economist Irving Fisher is also credited with the algebraically development of the theory. The quantity theory starts with the *transaction version* of the theory which is explained below: As per this version the following equation holds good:

$$M \times V = P \times T$$

Where,

M = Quantity of money i.e., money supply

V = Transaction velocity of money i.e., it shows how many times a dollar or a rupee changes hands in a period of time.

P = Price of a transaction

T = Number of the transactions in a period of time

The above equation is known as the transaction equation or quantity equation. To understand the meaning of the equation we consider the following hypothetical example:

Suppose the quantity of money is ₹ 25 and there are 50 transactions in a year with price of ₹ 5 per transaction, then as per the above equation:

25 V = 5 × 50 = ₹ 250

$$V = \frac{250}{25} = 10$$